

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA0307	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	B.Tech AI / DS	Date:	23/08/2026

Code of Conduct

I Siriyala Mubina[192324139] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S.Mubina

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Question 1 — Discourse, Reference Resolution & Text Coherence

Given paragraph: "John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

(a) Referring Expressions and Possible Antecedents

Mention	Type	Possible Antecedents
He	Pronoun (masc., sing.)	John
She	Pronoun (fem., sing.)	Mary
it	Pronoun (neut., sing.)	the park, a ball
him	Pronoun (masc., sing.)	John
they	Pronoun (plural)	John and Mary; John, Mary, and the dog

(b) Applying Constraints to Resolve References

1. Gender and Number Agreement

- "He" and "him" → must be masculine, singular → eliminates Mary; candidate remains John.
- "She" → must be feminine, singular → eliminates John, the dog; candidate remains Mary.
- "it" → must be neuter, singular → eliminates John, Mary, the dog; candidates remain "the park" and "a ball".
- "they" → must be plural → eliminates any single entity; candidates are {John, Mary} or {John, Mary, dog}.

2. Recency (prefer the most recently mentioned compatible antecedent)

- "He" (sentence 2) → most recent masculine singular mention is "John" (sentence 1) → resolved to John.
- "it" (sentence 3) → most recent neuter singular mention is "a ball" (sentence 2), which is closer than "the park" (sentence 1) → resolved to "a ball".
- "him" (sentence 4) → most recent masculine singular mention is "He" = John (sentence 2) → resolved to John.

3. Semantic Compatibility (animacy and action suitability)

- "wanted to play with it" — the verb "play with" selects an inanimate, manipulable object → "a ball" fits; "the park" (a location) is semantically weaker as a direct object of "play with" → confirms "it" = the ball.
- "The dog chased him" — "chased" requires an animate target capable of running → confirms "him" = John (a location/object cannot be chased in this sense).

4. Coherence (consistent entity chains across the discourse)

- Once He = John is fixed in sentence 2, every later masculine pronoun ("him") stays mapped to John unless a stronger local cue overrides it — this keeps a single unbroken entity chain instead of switching referents mid-paragraph.
- "they" in the final sentence must be coherent with the set of entities that have been active participants in the narrated event (going to the park, playing). The dog is a chased participant, not an agent of "going home," so the coherent reading is {John, Mary} — though a broader reading including the dog is linguistically possible, narrative coherence favors the two human protagonists as the topic entities.

(c) Constraint Graph / Table — Variables, Domains, Eliminations

Variable	Domain (candidates)	Constraint Applied	Eliminated	Result
He	John, Mary	Gender agreement (masc.)	Mary	John
She	John, Mary	Gender agreement (fem.)	John	Mary
it	the park, a ball	Number/gender (neut.) → Recency → Semantic compat. (play with)	the park	a ball
him	John, the dog	Gender agreement (masc.) → Semantic compat. (animate, can be chased)	the dog (as chase-agent excluded by role)	John
they	{J,M}, {J,M,dog}	Number (plural) → Coherence (narrative topic entities)	{J,M,dog} as primary reading	John and Mary

Constraint graph (textual form — nodes are mentions, edges are candidate links, crossed edges are eliminated):

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He --agreement--> [John]           She --agreement--> [Mary]
it  --recency+semantic--> [a ball]  (park eliminated by recency+semantics)
him --agreement+semantic--> [John]
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they --number+coherence--> [John, Mary]

Coreference chains:

- Chain-1 (John): John -> He -> him
- Chain-2 (Mary): Mary -> She
- Chain-3 (ball): a ball -> it
- Chain-4 (group): John and Mary -> they

(d) Final Resolved Coreference Chains and Rewritten Paragraph

- Chain 1 — Person (John): John → He → him
- Chain 2 — Person (Mary): Mary → She
- Chain 3 — Object (ball): a ball → it
- Chain 4 — Group: John and Mary → they

Rewritten paragraph with explicit references:

"John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John and Mary went home."

(e) Justification of Constraint Priority and Effect of Relaxing Recency

Priority order used: Gender/Number Agreement → Semantic Compatibility → Recency → Coherence.

- Gender/number agreement is applied first because it is a hard grammatical filter — it eliminates candidates with zero cost and no ambiguity (e.g., "She" can never bind to "John").
- Semantic compatibility is applied next because verb selectional restrictions (animacy, "play with" vs. "chased") often resolve ambiguity even before considering distance — a semantically impossible antecedent should be excluded regardless of how recent it is.
- Recency is used as a tie-breaker among candidates that already survive the first two filters (e.g., choosing "a ball" over "the park" for "it") — it is a soft, statistical preference, not an absolute rule.
- Coherence is applied last, as a global check across the whole discourse, to make sure the local pronoun-by-pronoun decisions still produce one consistent entity chain rather than contradictory chains.

Effect of relaxing the recency constraint:

If recency is relaxed, "it" becomes genuinely ambiguous between "the park" and "a ball" once gender/number and (weak) semantic filtering leave both as plausible candidates in a looser reading. This would force the resolver to rely solely on semantic compatibility, and in cases where the semantic filter is not sharp (e.g., a verb that could plausibly apply to either candidate), the system would need external world knowledge or would produce an incorrect/ambiguous resolution, reducing coreference accuracy — this is exactly why real coreference resolution systems (e.g., Hobbs' algorithm, Centering Theory) weight recency/distance heavily as a strong default heuristic.

Question 2 — Dialog, Discourse Planning & Surface Realization

Conversation: User: "I have an important exam tomorrow but I'm not able to concentrate."

Required Dialog Act: Advise + Encourage.

(a) Discourse Planning Steps Before Generation

- Content determination: identify the entities to reuse from the user's utterance — "exam", "concentrate", "you" — so the response is grounded in the user's actual situation rather than generic advice.
- Dialog act selection: decide the response will combine two acts — Advise (actionable suggestion) and Encourage (affective/motivational support) — and sequence them logically.
- Discourse relation planning: choose a rhetorical relation to connect the advice to its rationale — either Cause-Effect ("do X, so that Y improves") or Elaboration (add supporting detail to the advice).
- Lexical/keyword planning: ensure at least two of {focus, break, confident} appear naturally in the chosen content plan.
- Politeness and register planning: select a warm, non-judgmental tone, avoid imperatives that sound harsh, and keep sentence count to 2–3.
- Consistency check: verify the planned content does not contain contradictory claims (e.g., not simultaneously telling the user to "take a long break" and "don't stop studying").

(b) Three Candidate Responses

Response 1

"Take a short break to reset your focus, then come back to your notes — a clear mind absorbs more in less time. You've prepared well, so trust yourself and stay confident."

Response 2

"It's normal to feel this way before an important exam. Try breaking your revision into small chunks with short breaks in between — it will help you regain focus and feel more confident."

Response 3

"Step away for a five-minute break and breathe — pushing through without pausing usually hurts focus. When you return, you'll study with a calmer, more confident mind."

(c) Evaluation and Best Response

Response	Entity Coherence	Discourse Relation	Keywords (≥ 2)	Length	Consistency	Politeness/Tone
R1	Yes (exam implied, focus)	Cause-Effect	focus, confident	2 sentences ✓	Consistent	Warm, positive
R2	Yes (exam, concentrate→focus)	Elaboration	break, focus, confident	2 sentences ✓	Consistent	Warm, validating
R3	Partial (implicit exam ref.)	Cause-Effect	break, confident	2 sentences ✓	Consistent	Warm, calm

Best response: Response 2.

Response 2 is selected because it satisfies all three required keywords (focus, break, confident) rather than the minimum two, explicitly validates the user's emotional state before advising (stronger encourage component), reuses "exam" directly for tighter entity coherence, and uses an Elaboration relation that smoothly adds actionable detail without sounding like a blunt command — making it the most coherent and complete match to the constraint set.

(d) Effect of Violating Two Constraints

Example: violating (3) keyword inclusion and (6) politeness/positive tone.

- Without required keywords (e.g., no mention of "focus", "break", or "confident"), the response becomes generic and loses its grounding in the discourse plan — it could apply to any stressed user, weakening perceived relevance and helpfulness.
- Without politeness/positive tone (e.g., "You should have started earlier, just concentrate harder"), the response acquires a blaming, discouraging quality that directly contradicts the required Encourage dialog act — the system would technically produce advice, but fail the affective goal of the conversation, likely increasing user anxiety instead of reducing it.
- Combined effect: the dialog becomes task-correct but relationally poor — a known failure mode in conversational agents where constraint satisfaction on content is achieved but user experience and trust degrade because tone and lexical grounding constraints were dropped.

Question 3 — Machine Translation, Discourse Structure, Predicate Logic & Interpretation

Source sentence: "The bank by the river flooded after the storm, but it was saved by quick action."

(a) Word Sense Disambiguation of "bank"

Resolved sense: bank = riverbank (the land alongside a river), NOT a financial institution.

Justification using constraints:

- Collocation/context constraint: the phrase "bank by the river" explicitly anchors "bank" spatially next to "river" — a financial institution is not naturally described as being "by the river" in this construction, whereas "riverbank" is a standard geographic collocation.
- Selectional restriction constraint: the predicate "flooded" selects for a physical, low-lying, water-adjacent entity as its subject — a financial institution (an organization/legal entity) cannot literally "flood" in the physical sense being described here, but a landform can.
- World-knowledge/semantic compatibility constraint: "saved by quick action" fits a physical rescue/flood-mitigation scenario (e.g., sandbagging, evacuation, reinforcement) consistent with a riverbank, rather than a financial bank being "saved," which would typically require different framing (e.g., a bailout, described differently).

(b) Predicate Logic Representation

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∃x ∃y ∃s ( riverbank(x) ∧ river(y) ∧ located_by(x, y) ∧
           storm(s) ∧ flood(x) ∧ after(flood(x), s) ∧
           save(x) ∧ cause(quick_action, save(x)) )

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Reading: There exists a riverbank x , a river y , and a storm s such that x is located by y , x flooded after storm s occurred, and x was saved, with the saving caused by quick action.

(c) Target Sentence Satisfying All Constraints

Simple English paraphrase:

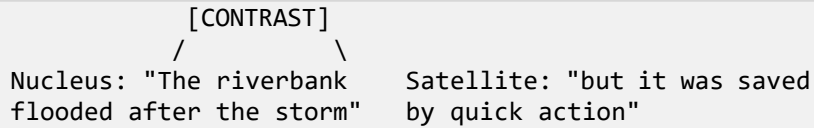
"The riverbank flooded after the storm, but quick action saved it."

Tamil paraphrase:

"புயலுக்குப் பிறகு ஆற்றங்கரை வெள்ளத்தில் மூழ்கியது, ஆனால் விரைவில் நடைபெற்ற அரங்க காப்பாற்றியது."

This paraphrase preserves the disambiguated sense (riverbank, not financial bank), keeps both entities (riverbank, storm) and the flood/save relations intact, and retains the Contrast relation via "but".

(d) Discourse Tree (RST-style)



Relation: CONTRAST

Nucleus : riverbank flooded (negative event)

Satellite: riverbank was saved (positive counter-event)

Connective: "but" signals the contrastive rhetorical relation.

(e) Advantages of Constraint-Based Approach over Pure Statistical MT

- Correct sense selection: a constraint-based approach explicitly encodes selectional restrictions ("flood" needs a physical location), so it reliably picks "riverbank" over "financial bank" even for a low-frequency sense — pure statistical MT can default to the higher-frequency sense (financial bank) if the training corpus is skewed, producing a mistranslation.
- Discourse-level accuracy: constraint-based methods explicitly preserve rhetorical relations (Contrast via "but"), ensuring the target sentence keeps the same logical structure — purely statistical (especially older phrase-based) systems translate more locally and can lose or blur discourse connectives across longer sentences.
- Explainability and control: each translation decision (WSD choice, relation preserved, entities retained) is traceable to an explicit rule/constraint, which is valuable for debugging and for domain-critical translation (legal, medical) — statistical/neural MT is comparatively a black box, making systematic errors harder to diagnose and fix.
- Trade-off (for balance): constraint-based systems require more manual linguistic engineering and generalize less gracefully to unseen constructions than data-driven neural MT, which is why modern systems often hybridize — using constraints/rules for disambiguation and discourse structure on top of neural fluency.

— End of Answer Script —